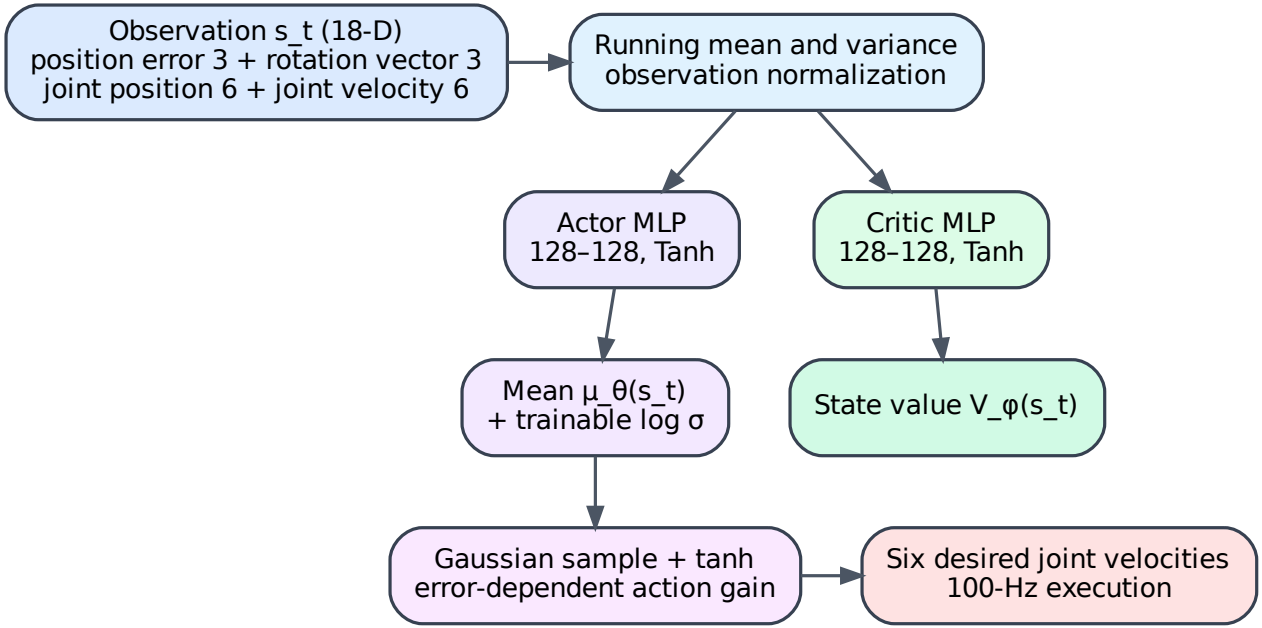
Observation s_t (18-D)
position error 3 + rotation vector 3
joint position 6 + joint velocity 6
Running mean and variance
observation normalization
Actor MLP
128–128, Tanh
Critic MLP
128–128, Tanh
Mean μ_θ(s_t)
+ trainable log σ
State value V_φ(s_t)
Gaussian sample + tanh
error-dependent action gain
Six desired joint velocities
100-Hz execution